

History of dentistry

Dentistry definition:

- Defined as evaluation , diagnosis , prevention , and / or treatment of diseases , disorders and /or conditions of the oral cavity , maxillofacial area and / or the adjacent and associated structures and their impact on the human body ;
- provided by **dentist** , within the scope of his / her education, training ,experience, in accordance with the ethics of the profession and applicable law.(as adopted by the 1997 ADA house of delegates)

History of dentistry

A profession that is ignorant of its past experiences has lost a valuable asset because “it has missed its best guide to the future.”

B.W. Weinberger

Dentistry: An Illustrated History (Mosby, 1995)

History of dentistry

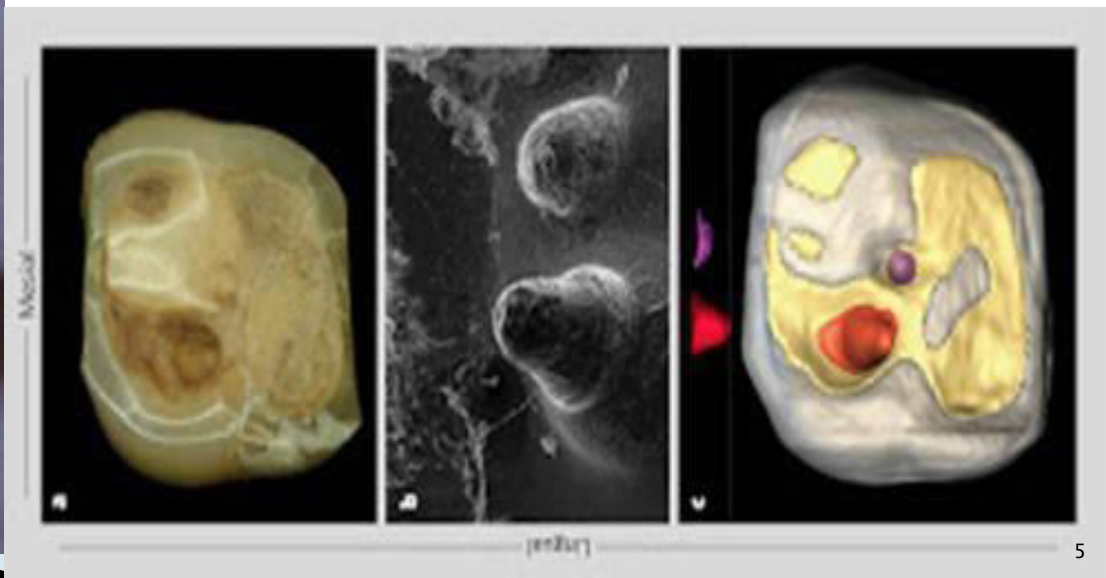
Ancient origins:

THE INDUS VALLEY CIVILIZATION

The Indus Valley Civilization has yielded evidence of dentistry being practiced as far back as 7000 BC. This earliest form of dentistry involved curing tooth related disorders with **bow drills** operated, perhaps, by skilled bead craftsmen. The reconstruction of this ancient form of dentistry showed that the methods used were reliable and effective.



History of dentistry



History of dentistry

THE SUMERIANS

A Sumerian text from 5000 BC describes a "tooth worm" as the cause of dental caries.

Evidence of this belief has also been found in ancient India, Egypt, Japan, and China.

The legend of the worm is also found in the writings of Homer, and as late as the 1300s AD the surgeon Guy de Chauliac still promoted the belief that worms cause tooth decay.



History of dentistry

The Egyptians

- ▶ 2600BC.....Death of **Hesy-re**, an Egyptian ,often called the (first dentist).An inscription of his tomb includes the title (**the greatest of those who deal with teeth, and of physicians**) this is the earliest known reference to a person identified as a dental practitioner.



History of dentistry

The Egyptians

▶ 2000 BC

“The earliest evidence of ancient dentistry we have is an amazingly detailed dental work on a mummy from ancient Egypt that archaeologists have dated to 2000 BC. The work shows intricate gold work around the teeth. This mummy was found with two donor teeth that had holes drilled into them. Wires were strung through the holes and then around the neighboring teeth.”



History of dentistry

The Egyptians

- ▶ 1700–1550 BC....An Egyptian text, the **Ebers papyrus**, refers to diseases of the teeth and various toothache remedies.



History of dentistry

THE CHINESE

By 2000 B.C. the Chinese were practicing dentistry.



History of dentistry

The Greeks

- ▶ 460-377 BC...**Hippocrates(father of medicine)** and **Aristotle (384-322 BC)** write about dentistry, including the eruption pattern of teeth, treating decayed teeth and gum disease, extracting teeth with forceps, and using wires to stabilize loose teeth and fractured jaws.
- ▶ **25 B.C.- 50 A.D. ...Aulus Celsus** roman medical writer, writes extensively in his important compendium of medicine on oral hygiene, stabilization of loose teeth, and treatments for toothache, teething pain, and jaw fractures.

History of dentistry

▶ The Etruscans

- ▶ 900–27 BC---The Etruscans practice dental prosthetics using gold crowns and fixed bridgework (bands and wires).



History of dentistry

THE ARABS

They were interested in the care of teeth rather than in their extraction and replacement, and mouth hygiene was a well established technique. They used small wooden stick, the end of which was often chewed, the wood fibers being used as a brush called "Siwak"



History of dentistry

The Middle Ages

Albucasis (936-1013) أبو القاسم الزهراوي

his medical encyclopedia in 30 volumes, called *al-Tasrif*, was translated into Latin in the twelfth century and was the medical text used in European universities until the seventeenth century. **The contributions of Albucasis to dentistry and periodontology were outstanding achievements.** He had a clear understanding of the major etiologic role of calculus deposits and described the techniques of scaling the teeth, using a set of instruments that he developed, splinting loose teeth with gold wire, and filing gross occlusal abnormalities.

History of dentistry

The Middle Ages

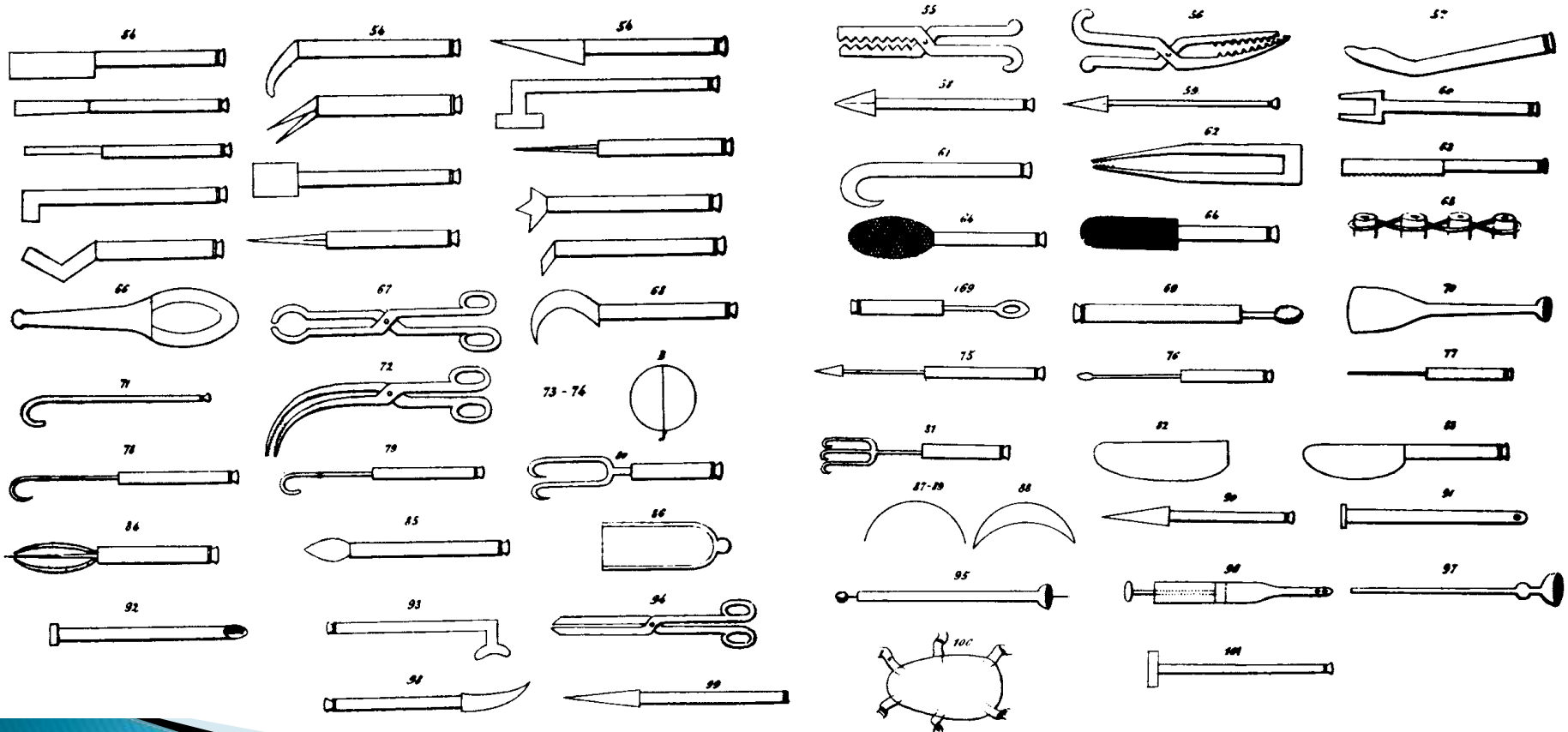
وربما كان اول من طبق نظرية "اخر العلاج الخلع" ففي كتاب التصريف يقول: "ينبغي أن تعالج الضرس من وجعه بكل حيلة وتتوانى عن قلعه، فليس منه خلف إذا قلع لأنه جواهر شريف، حتى إذا لم يكن بد من قلعه، فينبغي إذا عزم العليل على قلعه أن تتثبت حتى يصح عندك الضرس الوجع، فكثيراً ما يخدع العليل الوجع ويظن أنه في الضرس الصحيح فيقلعه ثم لا يذهب الوجع حتى يقلع الضرس المريض".



History of dentistry

The Middle Ages

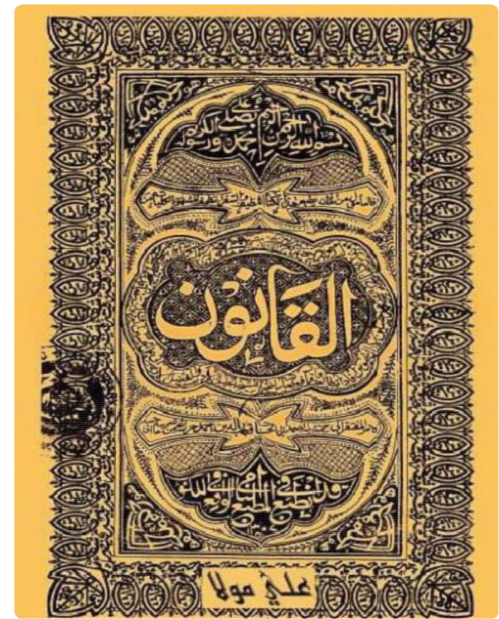
Illustration of Albucasis' periodontal instruments, showing scalers (sc), files (f), and the wiring of loose teeth (w).



History of dentistry

The Middle Ages

Avicenna (980–1037) (الشيخ الرئيس) was possibly the greatest of the Arabic physicians. His ***Canon***, *a comprehensive treatise on medicine*, was in continuous use for almost 600 years. Avicenna used an extensive "medical Materials" for oral and periodontal diseases and rarely resorted to surgery.



History of dentistry

The Middle Ages

- ▶ 700...AD. **A Medical text in china** mentions the use of (silver paste) a type of amalgam.
- ▶ 500–1000 AD....during the early middle ages in Europe – medicine, surgery, and dentistry are generally practiced by **monks**, the most educated people of that period.

History of dentistry

The Middle Ages

- ▶ 1130–1163...A **Series of papal edicts** prohibit monks from performing any type of surgery, bloodletting or tooth extraction. Barbers often assisted monks in their surgical ministry because they visited monasteries to shave the heads of monks and the tools of barber trade (sharp knives and razors...) were useful for surgery.
- ▶ **After the edicts**, Barbers assume the monks surgical duties: bloodletting, lancing abscesses, extracting teeth, etc.

History of dentistry

The Middle Ages

- ▶ 1210....**A Guild of barbers** is established in France. Barbers eventually evolve into two groups:
 - * **surgeons** who were educated and trained to perform complex surgical operations.
 - * **lay barbers**, or (**barber surgeons**), who performed more routine hygienic service including shaving, bleeding and tooth extraction.
- 1400...A Series of royal decrees in France prohibit lay barbers from practicing all surgical procedures except bleeding, cupping, leeching, and extracting teeth.

History of dentistry

The Renaissance

- ▶ 1530..AD – In this year the little medicinal book for all kinds of diseases and infirmities of the teeth published by (**Artzney Buchlein**), **the first book devoted entirely to dentistry**, is published in Germany. written for barbers and surgeons who treat the mouth, it covers practical topics such as **oral hygiene, tooth extraction, drilling teeth and placement of gold filling** .

History of dentistry

The Renaissance

- ▶ 1575–In France **Ambrose Pare**, known as the **father of surgery**, published his complete works. this includes practical information about dentistry such as tooth extraction and the treatment of tooth decay and jaw fractures.

History of dentistry

▶ The development of profession–18th century

- ▶ 1723–Pierre Fauchard, a French surgeon publishes:(**The Surgeon Dentist**), treatise on teeth(le chirurgien dentiste). **Fauchard is credited as being the father of modern dentistry** because his book was the first to **describe a comprehensive system for the practice of dentistry** including basic oral anatomy and function, operative and restorative techniques, and denture construction.



History of dentistry

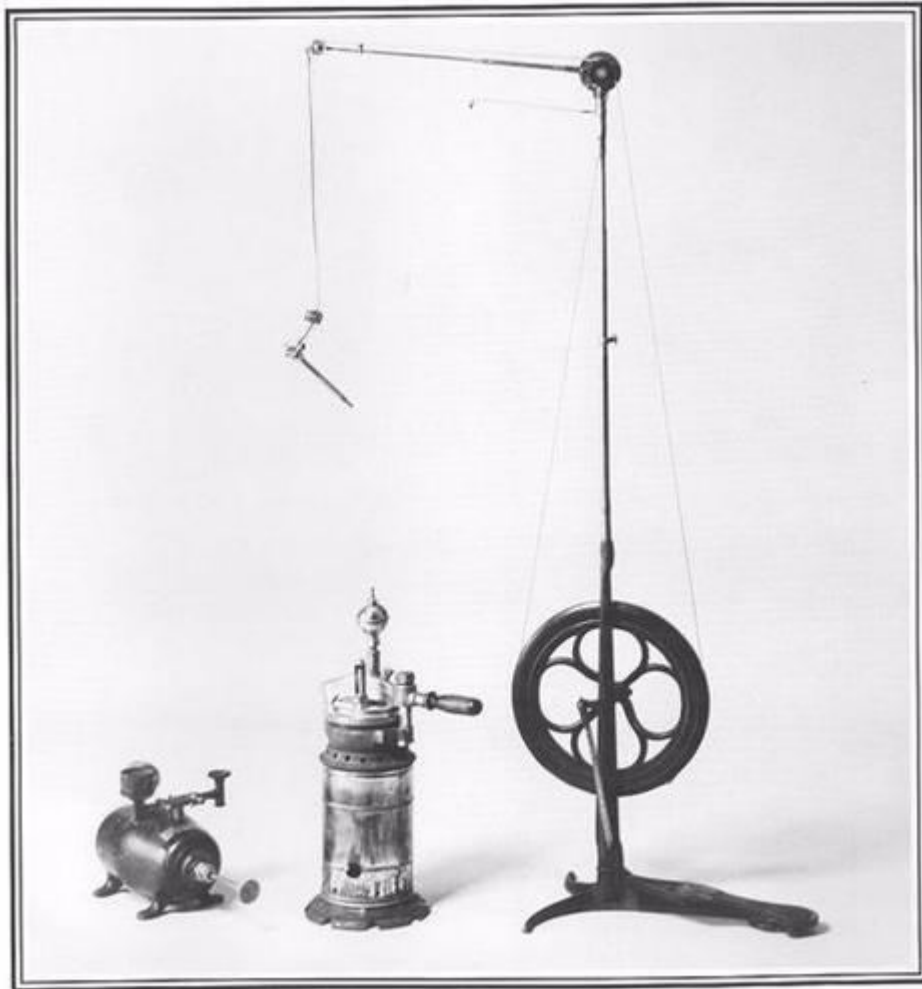
- ▶ **1746–claude mouton:** Describes a gold crown and post to be retained in root canal. He also recommends white enameling for gold crowns for a more esthetic appearance.



- ▶ **1760–John Baker:** The earliest medically-trained dentist to practice in America, immigrates from England and sets up practice.

- ▶ **1735–1818....Paul Revere:** In 1768 he Places advertisements in a Boston newspaper offering his services as a dentist. In 1776,in the first known case of post–mortem dental forensics, Revere verifies the death of his friend,Dr.Joseph warren in the Battle of Breed's Hill, when he identifies the bridge that he constructed for warren.
- ▶ **1789–Frenchman Nicolas Dubois De Chemant** receives the first patent for porcelain teeth.

- ▶ **1790–John Greenwood, one of George Washington’s dentists, constructs the first known dental foot engine.** He adapted his mother’s foot treadle spinning wheel to rotate a drill.



- ▶ **1790–Josiah Flagg**, a prominent American dentist, constructs the first chair made specifically for dental patients. To a wooden Windsor chair, **Flagg attaches an adjustable Headrest, plus an arm extension to hold instruments.**



Advances in science and education – 19th century

- ▶ 1801–**Richard c. skinner** writes the treatise on the Human Teeth, the first dental book published in America.
- ▶ 1825–**Samuel Stockton** begins commercial manufacture of porcelain teeth. His **S.S WHITE Dental Manufacturing Company** establishes and dominates the dental supply market throughout the 19th century.

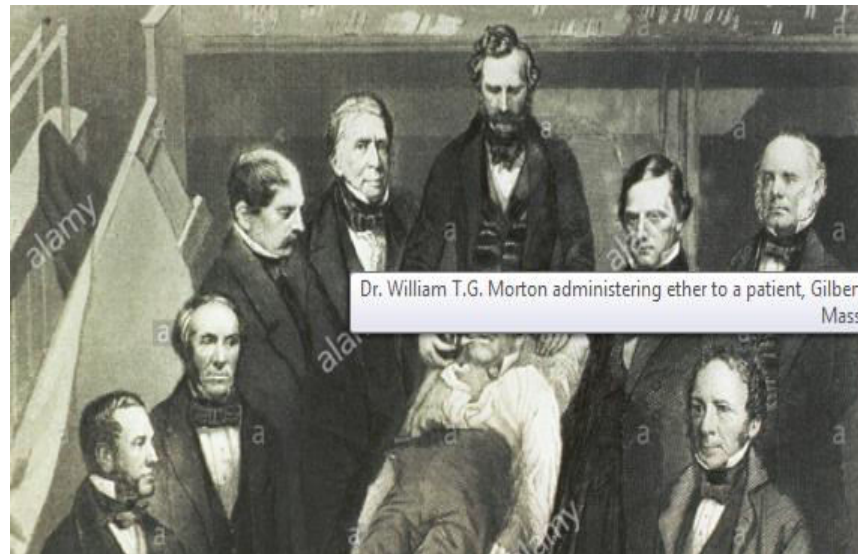
- ▶ 1832–James Snell Invents the first reclining dental chair.



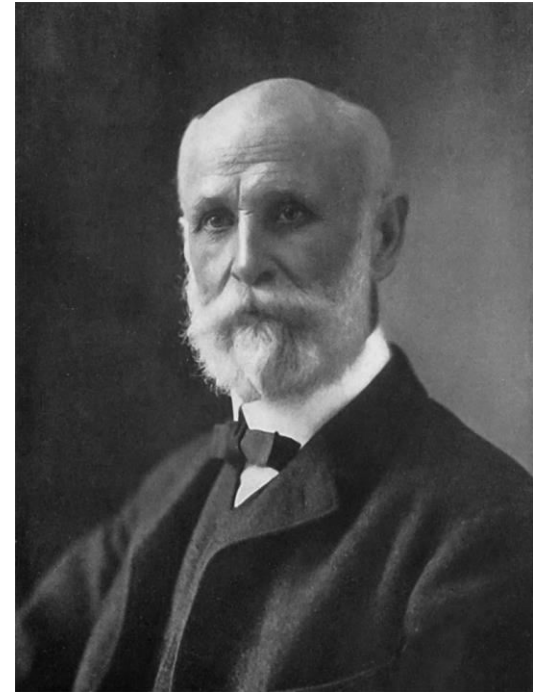
- ▶ 1833–1850: The **Crawcours** (Two brothers from France) introduce amalgam filling material in the united states under the name (Royal Mineral Succedaneum).
- ▶ 1839–The **American Journal Of Dental Science**, the world's first dental journal, begins publication.

- ▶ **1840–Horace Hayden and Chapin Harris**, establish the world's first dental school, the **Baltimore college of dental surgery**, and originated the doctor of dental surgery(DDS) degree.(the school merges with the university of **Maryland school of dentistry** in 1923).
- ▶ **1844– Horace wells**, discovers that nitrous oxide (laughing gas) can be used as an anesthesia and successfully uses it to conduct several extraction in his private practice. He conducts the first public demonstration of its use as an anesthetic in 1845 but the demonstration is generally considered a failure after the patient cries out during the operation.

- ▶ In 1846, another dentist (and a student of wells), **William Morton**, takes credit for the discovery when he conducts the first successful public demonstration of the use of ether as an anesthesia for surgery.



- ▶ **Greene Vardiman Black (1836–1915), commonly known as G.V. Black, is known as one of the founders of modern dentistry in the United States. He is also known as the father of operative dentistry.**
- ▶ He researched many important topics to dentistry, including the **cause of dental fluorosis** and **ideal cavity preparations**. One of his many inventions was a **foot-driven dental drill**. He is also known for his **principles of tooth preparations**. He organized '**Black's Classification of Caries Lesions**' which is still in use today.



- ▶ **1864–Sanford C. Barnum**, develops the **rubber dam**, a simple device made of a piece of elastic rubber fitted over a tooth by clamps, which solves the problem of isolating a tooth from the oral cavity.

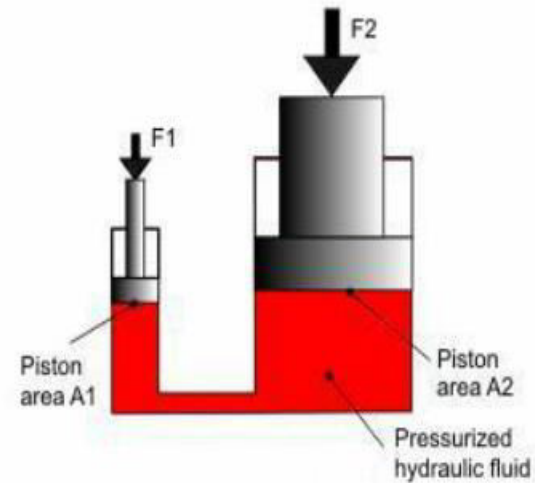
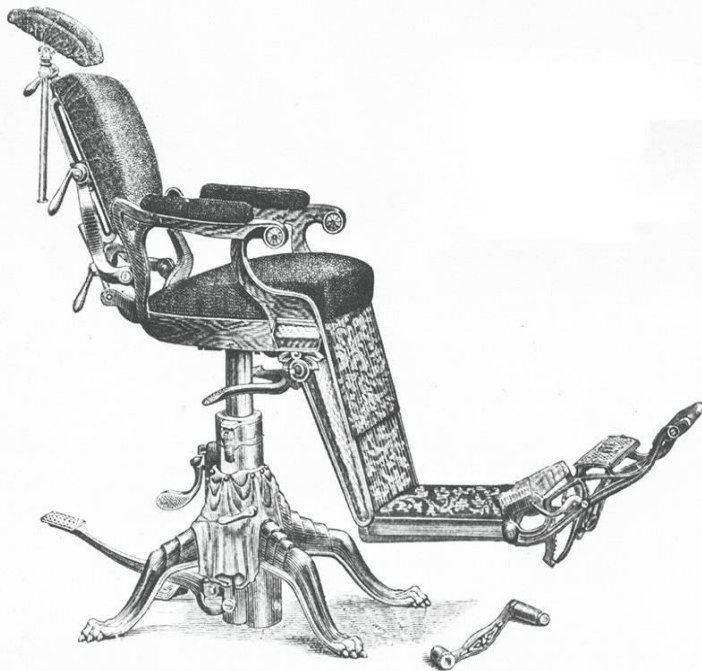


- ▶ 1866– Lucy HobbsTaylor, graduates from Ohio college of dental surgery, becoming the first women to earn a dental degree.

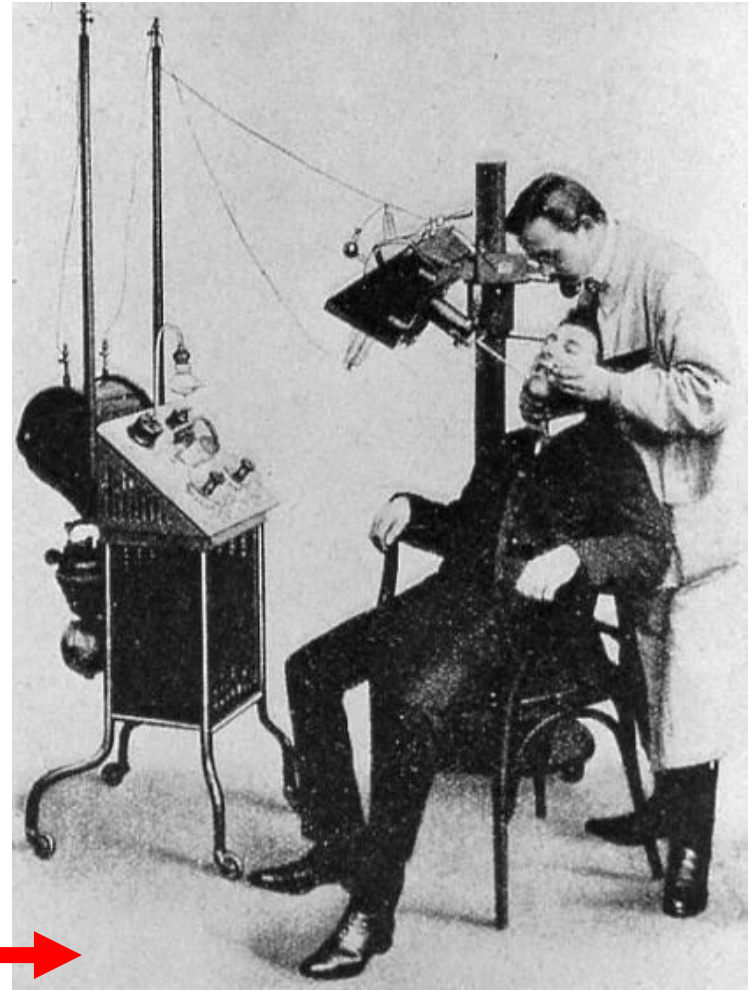
THE US'
FIRST
WOMAN
IN
DENTISTRY



- ▶ **1877–The Wilkerson chair, the first pump-type hydraulic dental chair, is introduced.**



- ▶ **1890-Willoughby Miller**, an American dentist in Germany, notes the microbial basis of dental decay in his book **Micro-organisms of the Human Mouth**. This generates an unprecedented interest in oral hygiene and starts a world-wide movement to promote regular tooth brushing and flossing.
- ▶ **1895-Wilhelm Roentgen**, a German physicist, discovers the x-ray
- ▶ **in 1896** prominent new Orleans dentist **C. Edmond Kells** takes the first dental x-ray of a living person in the U.S. →



- ▶ **1899–Edward Hartley Angle**, classifies the various forms of malocclusion. Credited with making orthodontics a dental specialty, **Angle** also establishes the first school of orthodontics.(Angle School of orthodontics in St.louis, 1900)

Innovations in techniques and technology– The 20th Century

- ▶ 1903– Charles land devised porcelain jacket crown.



- ▶ 1905–Alfred Einhorn, a German chemist, formulates the local anesthetic Procain, later marketed under the trade name Novocain.



- **1937–Alvin Strock**, inserts the first vitallium dental screw implant.
- **Vitalium**, the first successful biocompatible implant metal, had been developed a year earlier by **Charles Venable**, an orthopedic surgeon.



- **1938–The nylon tooth brush** the first made with synthetic bristles, appears on the market

- ▶ **1945– the water fluoridation** era begins when the cities of Newburgh, new York, and Grand rapids, Michigan, add sodium fluoride to their public water system.

- ▶ 1950– the first **fluoride toothpastes** are marketed
- ▶ 1958–A **fully reclining dental chair** is introduced



- ▶ **1960s – sit down four-handed dentistry** becomes popular in the U.S. this technique improves productivity and shortens treatment time



- ▶ 1960s–**lasers** are developed and approved for soft tissue procedures.
- ▶ 1960– The first commercial **electric tooth brush**, developed in **Switzerland**, is introduced in the united states. A Cordless, rechargeable model follows in 1961.

- ▶ 1989–The first commercial home tooth bleaching product is marketed
- ▶ 1990s– New tooth-colored restorative materials plus increased usage of bleaching, veneers, and implants start an era of esthetic dentistry.

Types of dentists

- All the dentists are general dentists, when they graduate from dental school
- Some dentists complete a general practice residency of 1 year or 2 to enhance their skills.
- Most of the typical needs can be well satisfied by a general dentist or general practitioner(GP) with DDS or BDS